Transit Analysis of Qatar 1-B with TASTE and TESS Observations

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Abstract

Qatar-1b, the first exoplanet discovered by the Qatar Exoplanet Survey, orbits a K3 spectral-type dwarf star on a circular path with an estimated orbital period of 1.42 days. This system is characterized by a planetary radius of approximately 1.136 Jupiter radii. Utilizing data from The Asiago Search for Transit Timing Variation of Exoplanets (TASTE) and NASA's Transiting Exoplanet Survey Satellite (TESS), our study employs data correction techniques, light curve analysis, and Markov Chain Monte Carlo (MCMC) simulations to refine the orbital and planetary parameters of Qatar-1b. We found the period of Qatar-1b to be consistent with previous studies, further confirming the planetary radius, semi-major axis, and inclination. Our analysis also included a detailed examination of limb darkening coefficients and the impact of transit timing variations.

1 Introduction

A hot Jupiter with a planetary companion on a nearby orbit would constitute an unprecedented planetary configuration, important for theories of the formation and evolution of planetary systems. The transiting planet Qatar-1 b is the first planetary body discovered by the Qatar Exoplanet Survey orbiting an old star with an age larger than 4 Gyr. It is a metal-rich dwarf star with a K3 spectral type. One confirmed planet was discovered around this star[1]. It is a typical hot Jupiter on a 1.42 d circular orbit [2] give a mass of $1.09 \pm 0.08\ M_{\rm Jup}$ (Jupiter's mass), but radial velocity (RV) measurements [4] show that the planet has a greater mass of $1.33 \pm 0.05\ M_{\rm Jup}$ [9] and its radius is $R_p \approx 1.136\,R_J$. It moves on a circular orbit with an orbital inclination $i_b = 84^\circ.26$, and the semi-major axis is $\frac{a_b}{R_*} = 6.319$ [9]. Covino et al [3] derived more accurate orbital parameters and spin-orbit alignment for the system. They also found the Qatar-1 to be a chromospherically active star. The first Transit Timing Variations (TTV) analysis of the system was carried out by Von Essen et al. [17]. They claimed that there are possible TTVs for Qatar-1b and speculated that the 190-day TTVs can be caused by a weak perturbator in resonance with Qatar-1b or by a massive body similar to a brown dwarf. However, the follow-up TTV studies by Maciejewski et al. [3], and Collins et al. [3] did not find any evidence of additional planets in the system. In contrast, Püsküllü et al. [16] found weak evidence of TTVs based on their later transit timing analysis, making Qatar-1b an interesting target for further study. Our study of Qatar-1b is based on data acquired by The Asiago Search for Transit Timing Variation of Exoplanets (TASTE) [13] at the Asiago Astrophysical Observatory in Italy and data from TESS. TESS is a NASA mission aimed at discovering transiting exoplanets. The mission was launched on April 18, 2018, and has monitored the brightness of more than 200,000 stars with a 4-CCD space telescope. TASTE, which started in 2009, intends to find Earth-sized planets using ground-based, medium-class telescopes. The TASTE data is observed during the transit event, while the TESS data set is multiple transits with a longer observation window, allowing for a period estimation. On that account, we first performed the analysis on the TESS data and used the resulting period estimation as a prior for the TASTE analysis. Several stellar parameters were estimated by Mislis et al., 2015 [11] with a spectroscopic analysis. These parameters were used as priors on our sampler. We used a Markov Chain Monte Carlo (MCMC) sampler to find the orbital and planetary parameters of Qatar-1b; the standard value of parameters is given in Table 1. In the following sections, we describe the procedures for correcting the TASTE (Section 2) and TESS (Section 3) data sets to obtain the final light curves and determining estimates of limb darkening coefficients, the orbital period and transit times which were used as priors and boundaries for the MCMC computation. Section 5 describes the Transit Modelling; the results can be found in Section 6.

2 TASTE

2.1 Data Correction

The TASTE data consists of 521x237 pixel images. We worked with 30 bias frames, 30 flat frames and 1857 science frames. The observations were performed on February 13th 2023, using the Copernico 1.82m reflecting telescope at the Asiago Astrophysical Observatory. The science images were taken with a 5.00-second exposure time.

2.1.1 Bias

When the exposure time is set to 0, as no photons fall on the CCD, the expected value for each pixel is 0. However, the baseline level of signal is recorded by the CCD when no light is falling on it. This signal is an inherent feature of the CCD and is caused by various factors, including the electronic properties and imperfections of the device. The bias frame maps the inhomogeneities subtracted from the image taken with the CCD. This frame is taken with a closed shutter to make the photon count zero. A master bias frame was obtained by stacking all 30 bias frames such that the final image contains the median value of each pixel. The error associated with this frame was calculated by selecting pixels between 300 and 350 on the x-axis to avoid noise near the edges and computing the standard deviation (STD) of all pixel values and is then calculated using the formula

$$\sigma_{D'} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(D'_i - \mu_{D'})^2}$$

where n is the number of elements in D' or the selected pixels, and $\mu_{D'}$ is the mean of D', defined as

$$\mu_{D'} = \frac{1}{n}\sum_{i=1}^{N} D'_i.$$

For the standard deviation of the selected pixel, we divide $\sigma_{D'}$ by the square root of the number of images N. This gives us the expected standard deviation of the median bias, $\sigma_{\text{median\_bias}}$, which is expressed as

$$\sigma_{\text{median\_bias}} = \frac{\sigma_{D'}}{\sqrt{N}}.$$

Combining these steps, we get the median bias.

$$\sigma_{\text{median\_bias}} = \frac{1}{\sqrt{N}}\sqrt{\frac{1}{n}\sum_{i=1}^{n}(D'_i - \mu_{D'})^2}.$$

2.1.2 Flat

Acquiring and processing flat frames begins with capturing a flat field image using a 1-second exposure time under uniform illumination with the telescope's CCD. This step helps identify and correct pixel sensitivity variations and illumination inhomogeneities. Each flat frame then has the master bias frame subtracted to eliminate electronic offset. To normalize the flat frames, the median value is calculated from a selected region (pixels 100 to 500 on the x-axis and 0 to 75 on the y-axis) where the illumination is most even, and all pixel values are divided by this median value. The error for each normalized flat frame, σ_{nf}, is computed using the formula

$$\sigma_{nf} = \sqrt{\frac{\sigma^2_{\text{median\_bias}} + \text{RON}^2 + F}{\text{Mdn}_{\text{flat}}}},$$

where RON is the read-out noise, F is the flux from the flat frame, and Mdn_{flat} is the median flux value of the flat frame. The normalized flat frames are then combined to create a master flat frame, representing the median value of each pixel across all frames. The error for the master flat frame, σ_{flat}, is calculated as

$$\sigma_{\text{flat}} = \sqrt{\frac{\sum_{\text{flat}} \sigma^2_{nf}}{N}},$$

where the sum is over all flat frames, and N is the total number of flat frames, with the median approximated by the average. This method assumes Poissonian statistics for photon flux and considers the error in the median value of each flat frame negligible.

2.1.3 Science Correction

The raw science frames are first corrected by subtracting the master bias frame, a process known as de-biasing. The error associated with the de-biased science frame (db) is given by

$$\sigma_{db} = \sqrt{\sigma^2_{bias} + \text{RON}^2 + db},$$

Table 1: Stellar and Planetary Parameters from Collins et al [3]

Parameter	Value
Stellar Parameters	
Mass (M)	$0.838^{+0.043}_{-0.041}\, M_\odot$
Radius (R)	$0.803 \pm 0.016\, R_\odot$
Luminosity (L)	$0.365^{+0.040}_{-0.034}\, L_\odot$
Density (ρ)	$2.286^{+0.074}_{-0.070}$ g/cm^3
Surface gravity ($\log g$)	$4.552^{+0.012}_{-0.011}$ cgs
Effective temperature (T_{eff})	5013^{+93}_{-88} K
Metallicity ([Fe/H])	$0.171^{+0.097}_{-0.094}$
Planetary Parameters	
Semi-major axis (a)	$0.02332^{+0.00040}_{-0.00038}$ AU
Mass (M_P)	$1.294^{+0.052}_{-0.049}\, M_J$
Radius (R_P)	$1.143^{+0.026}_{-0.025}\, R_J$
Density (ρ_P)	$1.076^{+0.057}_{-0.053}$ g/cm^3
Surface gravity ($\log g_P$)	3.390 ± 0.015 cgs
Equilibrium temperature (T_{eq})	1418^{+28}_{-27} K
Primary Transit Parameters	
Linear ephemeris from transits (T_0)	2456234.103218 BJD
Linear ephemeris period from transits (P)	1.4200242days
Planet-to-star radius ratio (R_P/R)	$0.14629^{+0.00063}_{-0.00064}$
Semi-major axis in stellar radii (a/R)	$6.247^{+0.067}_{-0.065}$
Inclination (i)	$84.08^{+0.16}_{-0.15}$ degrees
Impact parameter (b)	$0.645^{+0.010}_{-0.011}$
Transit depth ()	0.02140 ± 0.00019

where σ_{db} is the error in the de-biased science frame, σ_{bias} is the error in the master bias frame, RON is the readout noise of the detector, and db is the flux value in the de-biased science frame. After de-biasing, the science frame is divided by the master flat frame to correct for pixel-to-pixel variations in sensitivity, a step known as flat-field correction. The error associated with the corrected science frame (sc) is then calculated using

$$\sigma_{sc} = sc \cdot \sqrt{\left(\frac{\sigma_{db}}{db}\right)^2 + \left(\frac{\sigma_{flat}}{flat}\right)^2},$$

where σ_{sc} is the error in the corrected science frame, sc is the flux value in the corrected science frame, σ_{db} is the error in the de-biased science frame, db is the flux value in the de-biased science frame, σ_{flat} is the error in the master flat frame, and $flat$ is the flux value in the master flat frame. This two-step process ensures that the final error in the corrected science frame is accurately determined.

The errors for the TASTE data analysis were all computed using the error propagation equation:

$$\sigma_F = \sqrt{\left(\frac{\partial F}{\partial X_1}\sigma_1\right)^2 + \left(\frac{\partial F}{\partial X_2}\sigma_2\right)^2 + \left(\frac{\partial F}{\partial X_3}\sigma_3\right)^2 + \ldots},$$

which gives the error σ_F associated with a variable F that is a function of X_1, X_2, and $X_3, \ldots$ with respective errors $\sigma_1, \sigma_2, \sigma_3, \ldots$.

2.2 Aperture Photometry

Simple Aperture Photometry (SAP) measures the flux (light intensity) within a defined boundary around the target star on an image, creating a light curve that tracks the star's brightness over time. The total flux F_{ap} within an aperture is calculated by summing the flux of all pixels within a circular area centered on the star, with the sky background flux F_{sky} subtracted to get a corrected flux $F_{\text{ap,corr}}$.

An annular region around the star, with an inner radius of 10 pixels and an outer radius of 18 pixels, is used to estimate the sky background flux:

$$F_{\text{sky}} = \frac{\sum_{i=1}^{N_{\text{ann}}} F_i}{N_{\text{ann}}}$$

where N_{ann} is the number of pixels in the annular region. The corrected flux is then:

$$F_{\text{ap,corr}} = F_{\text{ap}} - N_{\text{ap}} \cdot F_{\text{sky}}$$

where N_{ap} is the number of pixels in the aperture. The error associated with the sky flux, σ_{sky}, is given by:

$$\sigma_{\text{sky}} = \sqrt{\frac{\sum_{i=1}^{N_{\text{ann}}} \sigma_{\text{sc},i}^2}{N_{\text{ann}}^2}}$$

where $\sigma_{\text{sc},i}$ is the error in the flux of the i-th pixel. The error in the aperture photometry flux, σ_{ap}, is:

$$\sigma_{\text{ap}} = \sqrt{\sum_{i=1}^{N_{\text{ap}}} \sigma_{\text{sc},i}^2 + N_{\text{ap}}^2 \cdot \sigma_{\text{sky}}^2}$$

Differential photometry improves the accuracy of the light curve by comparing the target star's flux to that of reference stars, reducing atmospheric and instrumental noise effects. The differential flux F_{dp} is:

$$F_{\text{dp}} = \frac{F_{\text{tg}}}{F_{\text{ref}}}$$

where F_{tg} and F_{ref} are the target and reference star fluxes, respectively (Fig 1). The error in the differential flux, σ_{dp}, is:

$$\sigma_{\text{dp}} = \sqrt{\left(\frac{\sigma_{\text{ap,tg}}}{F_{\text{ref}}}\right)^2 + \left(\frac{F_{\text{tg}} \cdot \sigma_{\text{ap,ref}}}{F_{\text{ref}}^2}\right)^2}$$

where $\sigma_{\text{ap,tg}}$ and $\sigma_{\text{ap,ref}}$ are the errors in the aperture photometry of the target and reference stars.

Choosing the optimal aperture radius involves balancing the inclusion of the star's light and minimizing the sky background. The flux inside a 10-pixel radius normalizes the flux inside different radii. A radius of 6 pixels is chosen, capturing about 95% of the total flux (Fig 2).

A second-degree polynomial given by:

$$P(t) = a_2 t^2 + a_1 t + a_0$$

is fitted to the out-of-transit data to correct the systematic trends, and then a normalized light curve is made F_{norm}

$$F_{\text{norm}} = \frac{F_{\text{dp}}}{P(t)}$$

The standard deviation (STD) of the out-of-transit flux is calculated to determine the most stable reference star. The STD of the normalized flux outside the transit is:

$$\sigma_{\text{out-transit}} = \sqrt{\frac{1}{N} \sum_{i=1}^{N} \left(F_{\text{norm},i} - \overline{F_{\text{norm}}}\right)^2}$$

Here, N is the number of out-of-transit data points, and $\overline{F_{\text{norm}}}$ is the mean normalized flux. The analysis reveals the standard deviations of the out-of-transit normalized light curves: 0.0052 for reference star 1 and 0.0039 for reference star 2, and the sum of the reference star giving 0.0038. Since the sum of the reference star has a lower value, we choose that for modelling. The normalized light curve provides a clear view of the transit event with minimized noise, enabling precise transit modelling (Fig 3).

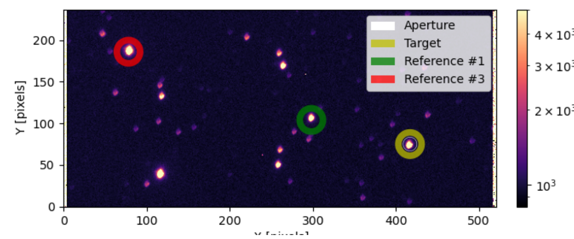

Figure 1: Science frame showing the annular selection of the sky around Qatar 1B and the 2 reference stars.

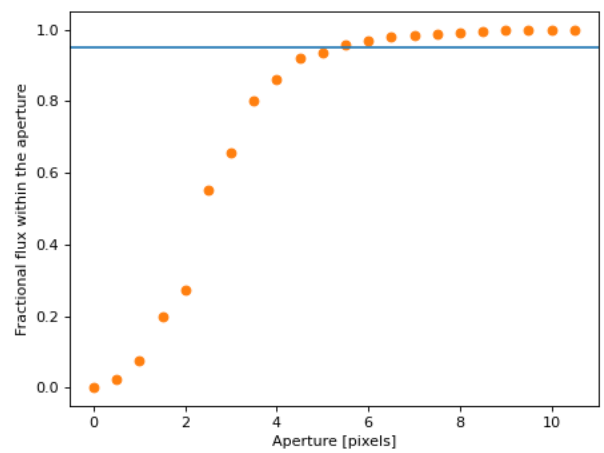

Figure 2: Flux from the target star inside the aperture selection for different radii, normalized by the flux inside an aperture of 10 pixels. The horizontal line indicates 95% of the total flux.

3 TESS

3.1 Data Flagging

TESS (Transiting Exoplanet Survey Satellite) performed short-cadence observations of the exoplanet QATAR-1b in several sectors. Data from three distinct sectors—59, 51, and 76—were selected from the MAST (Mikulski Archive for Space Telescopes) portal for this analysis. These observations were conducted on 18 January 2023, April 23, 2022, and May

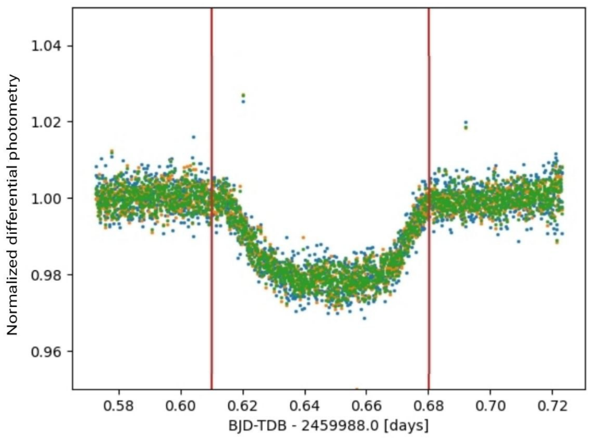

Figure 3: Normalized differential photometry of Qatar-1b showing transit event on the TASTE dataset

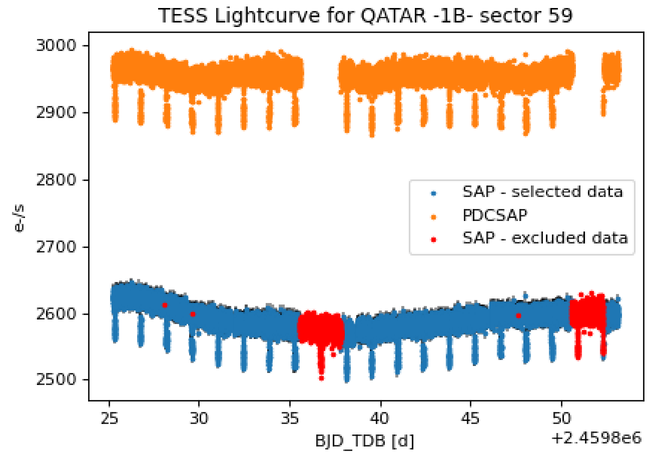

Figure 5: TESS observation sector 59 where the flagged data is marked as red

8, 2024. Each sector provided approximately one month of continuous observation, with each exposure lasting 120 seconds. TESS uses a red-optical bandpass covering the wavelength range from about 600 to 1000 nm. The TESS mission provides observational data in the form of Target Pixel Files (TPFs). These files are structured to store detailed information about each observed target star. TPFs consist of pixel data arranged in a grid format, with each pixel corresponding to a specific area on the CCD (Charge-Coupled Device). The aperture bitmask from the TPF file revealed the pixel coverage used for photometry.The SAP flux and PDCSAP flux for sector 59 are shown in Fig 5.

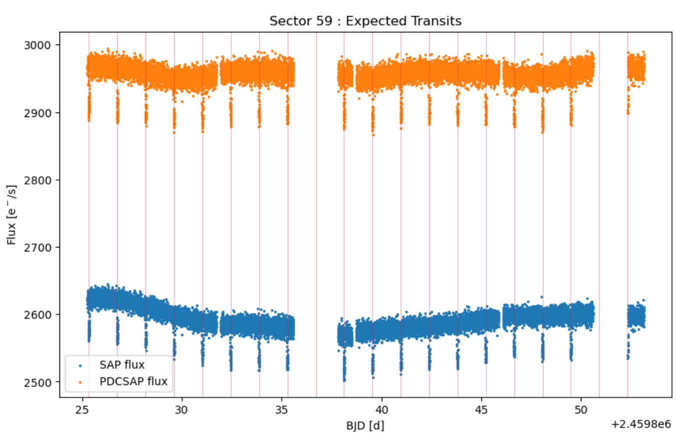

Figure 4: TESS SAP and PDCSAP of QATAR 1B from observation sector 59, with the estimated transit times

The TESS mission also detected the transit of QATAR-1b with a period of 1.4200242 $\pm$ 0.0000012 days, centered at the epoch 2456234.1032 $\pm$ 0.0003 BJD (Barycentric Julian Date). Based on these parameters, simulations were conducted to predict the expected times of subsequent transits during the observational period. These are indicated by vertical lines in Fig 4; the figure shows that the dips in the light curve coincide with the estimated transit times, supporting the reliability of TESS's estimation. The data provided by TESS contains a flag parameter, indicating data taken during an anomalous event. Figure 5 shows those excluded from this analysis.

3.2 Data De-trending

TESS provides two primary flux products: SAP (Systematic Aperture Photometry) and PDCSAP (Pre-search Data Conditioning SAP). SAP offers initial light curves directly from spacecraft data, while PDCSAP applies corrections for systematic errors, enhancing data reliability. We compared Simple Aperture Photometry (SAP) and Pre-search Data Conditioning SAP (PDCSAP) fluxes from TESS observations of Qatar 1b taken in sector 59. The SAP data were selectively plotted based on conservative quality criteria to exclude flagged observations. Error bars were included to visualize the uncertainty in SAP flux measurements. Variations in flux, including noise and flares, introduce complexities in the light curve, which can challenge the fitting of transit models that assume only transit-related flux variations. To address this, three different de-trending methods were tested.The PDCSAP already corrects for instrumental noises, yet additional flux variations persist in the data, such as intrinsic stellar activity or flares. Though, for the analysis we only used the SAP which showed a lower STD when combined with the Huberspline algorithm.Two methods utilized the

Python package Wotan [5] with different estimators: Tukey's biweight and Huber spline (Fig 6). Both methods apply weights to data points based on their distance from the center within a window function that divides the data into small intervals. The biweight method assigns zero weight beyond a specified cutoff distance, disregarding outliers. In contrast, the Huber spline method maintains non-zero weights throughout, creating a continuous function to describe the trend. The window length was selected to remove unwanted trends while preserving transit shapes, set at 0.5 BJDs, with a break tolerance of 0.4 BJDs and a tuning parameter of 3.0. For the biweight method, an additional edge cutoff of 0.1 BJDs was applied.The standard deviations (STDs) of the resulting data from each method were computed for the out-of-transit flux. This assessment aimed to determine the optimal method based solely on minimizing the STD. Table 2 presents the results for each sector.

For all sectors, the Huber spline method gives the best result. The data de-trended with this method was thus chosen for further analysis. The errors in the flux were divided by the trend accordingly, as Wotan does not include this conversion.

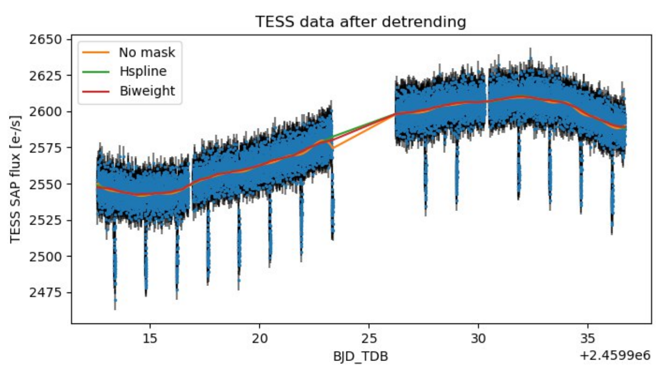

Figure 6: Comparison plot for the trend in PDCSAP, Biweight, and Huber spline methods.

The resulting normalized light curve provided a clear view of the relative changes in flux, allowing for accurate analysis and interpretation of transit events and other astronomical phenomena.

3.3 Period and Time of Transit Initial Estimations

To determine parameter boundaries for the period and epoch at the central time of transit for modelling purposes, we employed the TransitLeastSquares algorithm (TLS) [6]. This method utilizes transit-like functions to model the data, accounting for effects such as stellar limb darkening, planetary ingress, and egress.

The period, transit depth, and Transit duration were calulated using (TLS)[6], which uses transit-like functions to model the data, considering effects of stellar limb darkening and planetary ingress and egress. We used this algorithm on each sector of the TESS data to find the transit period of Qatar-1b, obtaining as a result 1.4197 days, 1.4205 days and 1.4194 days for sectors 59, 51 and 76, respectively. Through TLS, transit duration(days) was calculated to be 0.0533(Sector 59), 0.0485(Sector 51), and 0.0507(Sector 76).

4 Limb Darkening

Limb darkening is an optical phenomenon observed in stars, characterized by a decrease in specific intensity $I_\lambda(\mu)$ as one moves from the center to the limb of the star's visible disk. This effect arises due to the varying optical depth within the star's photosphere, causing photons emitted from deeper, hotter layers to dominate the observed brightness at the center relative to those emitted from cooler, outer layers at the limb.

The limb darkening coefficient $\mu = \cos(\theta)$, where θ is the angle between the line of sight and the surface normal, describes how the specific intensity of light $I_\lambda(\mu)$ varies across a star's visible disk.

To model stellar limb darkening the Limb Darkening Toolkit (LDTk) [14] was used, a Python package that estimates stellar limb darkening coefficients for a chosen model based on given stellar parameters and the filter used for the observation. The effective temperature, surface gravity and metallicity of Qatar 1B given as input were taken from Collin et al. [3]. These can be found in Table 1. For this analysis, we chose to use the quadratic model, which follows [14]:

$$\mathrm{I}(\mu) = 1 - \mathrm{C_u}(1-\mu) - \mathrm{C_v}(1-\mu)^2, \tag{1}$$

where $\mathrm{I}(\mu)$ is the stellar intensity as a function of $\mu = \sqrt{1-\mathrm{z}^2}$, where z is the normalized distance from the center of the stellar disk. The model is thus described by 2 coefficients, $\mathrm{U_1}$ and $\mathrm{U_2}$, which LDTk calculates. The resulting estimated values of these coefficients by LDTk are $\mathrm{U_1} = 0.527\pm0.017$ and $\mathrm{U_2} = 0.136\pm0.040$ for the r-Sloan band, used by TASTE for this observation and $\mathrm{U_1} = 0.47\pm0.02$ and $\mathrm{U_2} = 0.12\pm0.040$ for TESS filter.

Table 2: Standard deviation values of different de-trending methods for each TESS sector.

Method	STD		
	Sector 59	Sector 51	Sector 76
SAP Biweight	0.002630	0.002944	0.002932
SAP Huber spline	0.002624	0.002950	nan
PDCSAP biweight	0.002637	0.003047	0.003132
PDCSAP Huber spline	0.002634	0.003053	nan

Table 3: Parameters for each TESS sector from TLS.

Parameter	Sector		
	Sector 59	Sector 51	Sector 76
Transit Depth	0.97839	0.97921	0.97834
Transit Duration (days)	0.05332	0.04849	0.05069
Period (days)	1.41971	1.42053	1.41935
Number of Transits	18	17	18

5 Transit Modelling

For estimating the parameters concerning QATAR 1B and its orbit, `BATMAN` [8] was used. It creates a model of the light curve and calculates the orbital parameters and properties of the exoplanet. It takes the normalized light curve as input, the boundaries for the transit period and epoch values, and the priors for the stellar and orbital parameters. We chose to work with an MCMC sampler to fit the model. The orbit was assumed to be circular due to tidal circularization, which is usually true for gas giants in small orbits [12]. This choice also sets the argument of the periastron automatically to 90 degrees. We calculated the log-likelihood of a model fit to two datasets, TESS and TASTE, by incorporating the errors and additional noise parameters (jitter). First, the TESS and TASTE data errors are adjusted to include an additional noise parameter. The log-likelihood calculation begins by adjusting the errors for the TESS and TASTE datasets to include additional noise parameters, also known as jitter. For the TESS data, the adjusted error is computed as the sum of the squared original error and the square of the jitter parameter, θ_1. This is expressed as

$$\sigma^2_{\mathrm{TESS},i} = \sigma^2_{\mathrm{TESS,\ original},i} + \theta_1^2$$

where $\sigma_{\mathrm{TESS,\ original},i}$ represents the original error for the i-th data point in the TESS dataset. Similarly, for the TASTE data, the adjusted error includes the original error and the jitter parameter θ_2, given by

$$\sigma^2_{\mathrm{TASTE},i} = \sigma^2_{\mathrm{TASTE,\ original},i} + \theta_2^2$$

where $\sigma_{\mathrm{TASTE,\ original},i}$ is the original error for the i-th data point in the TASTE dataset.

The total number of data points, N, is determined by summing the lengths of the adjusted error arrays for both TESS and TASTE datasets. The chi-squared values for TESS and TASTE are then calculated by summing the squared differences between the observed and expected values, normalized by their adjusted errors. For TESS, this is expressed as

$$\chi^2_{\mathrm{TESS}} = \sum_i \frac{(y_{\mathrm{TESS},i} - f_{\mathrm{TESS},i})^2}{\sigma^2_{\mathrm{TESS},i}}$$

where $y_{\mathrm{TESS},i}$ and $f_{\mathrm{TESS},i}$ are the observed and expected values, respectively. For TASTE, the chi-squared value is

$$\chi^2_{\mathrm{TASTE}} = \sum_i \frac{(y_{\mathrm{TASTE},i} - f_{\mathrm{TASTE},i})^2}{\sigma^2_{\mathrm{TASTE},i}}$$

Additionally, the sums of the logarithms of the adjusted errors for both datasets are computed: $\sum \ln(\sigma^2_{\mathrm{TESS},i})$ for TESS and $\sum \ln(\sigma^2_{\mathrm{TASTE},i})$ for TASTE. Finally, the log-likelihood $\ln L$ was calculated by combining the chi-squared values for the TESS and TASTE datasets. It also incorporates the sums of the natural logarithms of the squared errors, providing a measure of data uncertainty.

In the analysis of the TASTE dataset, the observational data consisted of normalized differential photometry flux recorded as a function of BJD (Barycentric Julian Date).The transit model was set up, and the priors for the stellar parameters were the same as those provided for TESS and the boundaries provided for the period. The boundaries used for the transit epoch were the first and last BJD of the TASTE observations. Since the TASTE data does not contain multiple transits, no information can be obtained about the period based on the light curve alone. We, therefore, introduced a Gaussian prior for the planetary period, given by the result obtained with TESS. We also provided a prior for the limb darkening coefficients. The light curve used as input for the TESS analysis was the SAP flux as a function of the BJD, de-trended with the Huber spline method. The estimation was done for each TESS sector separately, and we selected sector 59, which gave less error. The errors generated for different sectors are mentioned in Table 2. The boundaries used for the orbital period were determined by 1.42 days obtained with TLS, ± 0.1 d. The boundaries used for the transit epoch were determined by the first transit epoch given by TLS for each sector, ± 1 BJD. The first epoch value from sector 59 was used. The prior stellar parameters used were the results by Collins et al. 2017. All priors were set as Gaussian distributions. The modelled data with the original and the residue is plotted in Fig 7 and Fig 8.

6 Results and Conclusion

In this section, we present our results and study their compatibility with previous works on Qatar-1b.

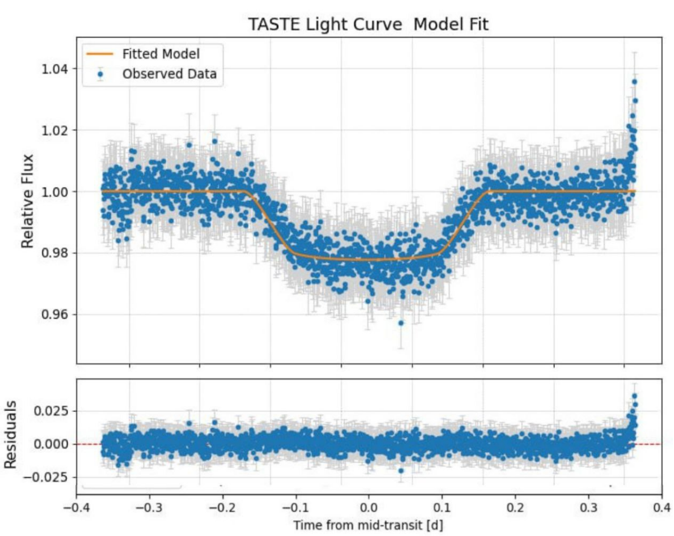

Figure 7: Modelled TASTE data against observed data

Notably, the transits observed by TESS were spaced over two years, potentially contributing to transit variations, as suggested by the results post-application of the Transit Least Squares (TLS) algorithms. Conversely, TESS sector 59 data was obtained in January 2023, and TASTE data was acquired in February 2023, with minimal temporal separation.

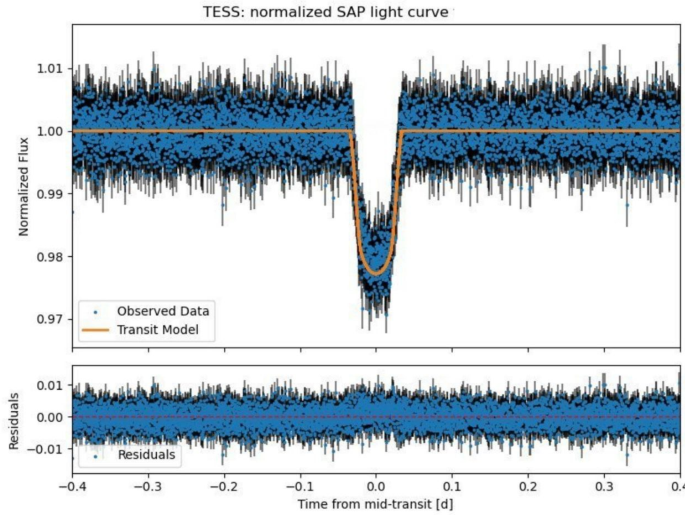

Figure 8: Modelled TESS fit against phase-folded data from TESS sector 59

Figures 7 and 8 present the transit models generated by `BATMAN` [8] for both TESS and TASTE datasets, alongside the observed flux post-application of filtering algorithms. The MCMC analysis results are illustrated in Figure 9 through corner plots, which demonstrate the correlations between the various parameters. These plots display well-defined peaks, indicating successful convergence of the model to distinct parameter values.

Table 4 enumerates all the parameters estimated by `BATMAN` for the full TESS and TASTE datasets. Parameters 1-9 were directly obtained from the model, while parameters 10-16 were derived based on input stellar properties from the prior. The analysis spanned three sectors, with sector 59 providing the most precise results due to the lowest standard deviation (0.0027) after applying the Huberspline algorithm to SAP flux values. Nonetheless, the results across all sectors were consistent with each other.

Previous estimations of the period (days) were $1.42002420^{+0.00000022}_{-0.00000022}$ by Collins et al. 2017 [3], $1.4200244443^{+0.000000049}_{-0.000000049}$ by Kokori et al. 2023 [7], $1.42002433^{+0.00000003}_{-0.00000003}$ by Mannaday et al. 2022 [10], $1.4200242^{+0.0000002}_{-0.0000002}$ by Patel & Espinoza 2022 [15], $1.4200246^{+0.0000004}_{-0.0000004}$ by von Essen et al. 2013 [17], and $1.42002504^{+0.00000071}_{-0.00000071}$ by Covino et al. 2013 [4]. All the published results agree with our estimation of the period of Qatar 1b within the error range. Our estimation for the radius of the planet in terms of stellar radius (R_p/R_s) came out to be $0.1270^{+0.0096}_{-0.0088}$, comparable with the results from von Essen et al.

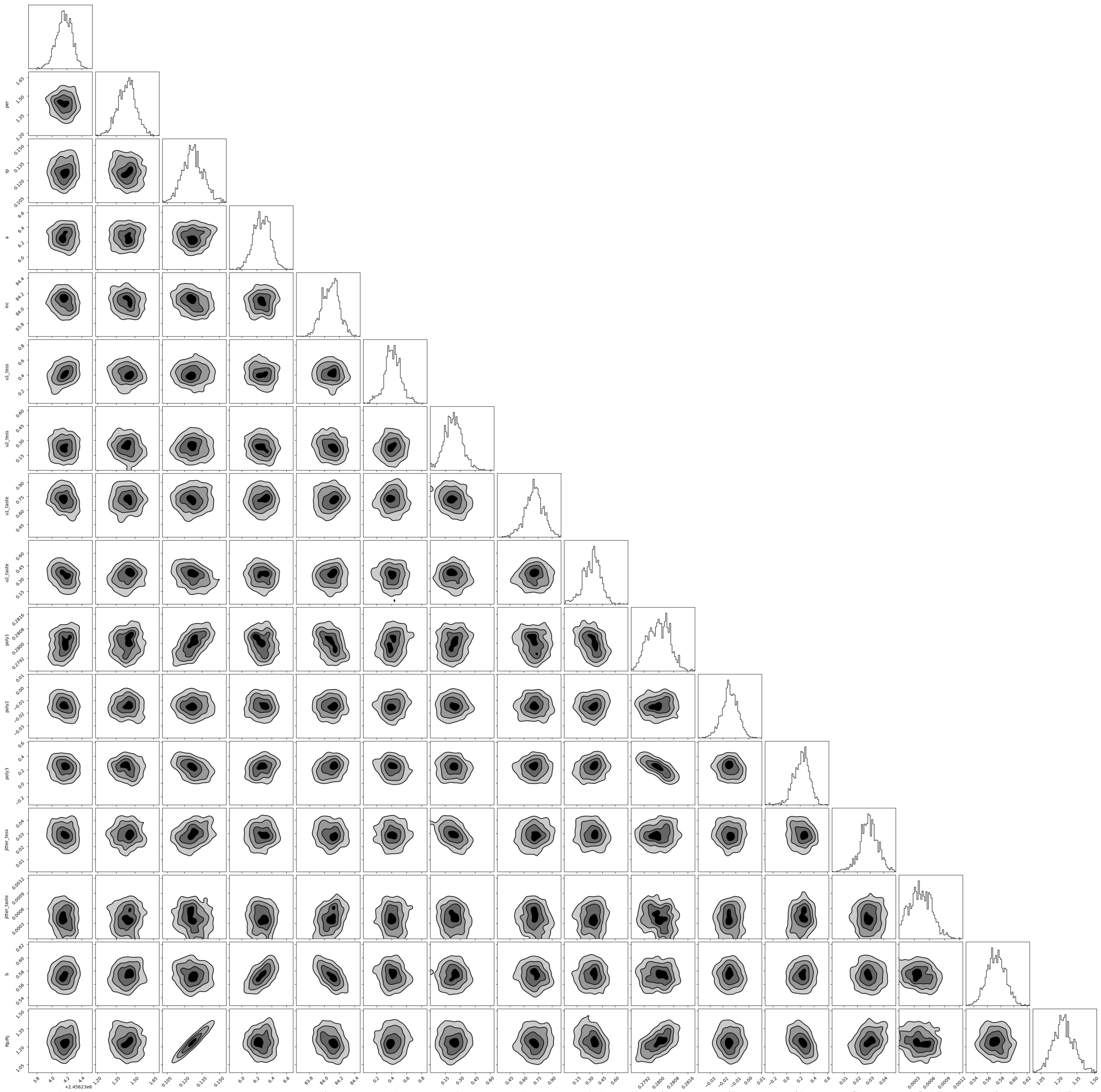

Figure 9: Corner plot of parameters from MCMC calculation. Central transit time (t_0), orbital period (per), planet-to-star radius ratio (r_p), semi-major axis to stellar radius ratio (a), orbital inclination (inc), first limb darkening coefficient (u1), second limb darkening coefficient (u2), first polynomial coefficient (poly1), second polynomial coefficient (poly2), third polynomial coefficient (poly3), additional noise parameter (jitter), impact parameter (b) and planetary radius in Jupiter units ($\frac{R_p}{R_j}$).

Parameter Description	Value	Uncertainty
t_0 Time of Transit Center (Julian Date)	2456234.1664255	$^{+0.1003616}_{-0.1049465}$
P Orbital Period (days)	1.4391 days	$^{+0.0663}_{-0.0733}$
R_p (planet radius (in units of stellar radii))	0.1270	$^{+0.0096}_{-0.0087}$
a Semi-major Axis (in units of stellar radii)	6.2715	$^{+0.1086}_{-0.1128}$
i Inclination (degrees)	84.0887°	$^{+0.1022}_{-0.1202}$
$u1_{\text{TESS}}$ Limb-darkening Coefficient (TESS)	0.4119	$^{+0.1039}_{-0.0936}$
$u2_{\text{TESS}}$ Limb-darkening Coefficient (TESS)	0.2303	$^{+0.0854}_{-0.0866}$
$u1_{\text{TASTE}}$ Limb-darkening Coefficient (TASTE)	0.7126	$^{+0.0969}_{-0.0971}$
$u2_{\text{TASTE}}$ Limb-darkening Coefficient (TASTE)	0.3371	$^{+0.0860}_{-0.1109}$
poly1 Polynomial Coefficient 1	0.28000	$^{+0.00058}_{-0.00066}$
poly2 Polynomial Coefficient 2	-0.0147	$^{+0.0063}_{-0.0061}$
poly3 Polynomial Coefficient 3	0.2273	$^{+0.0995}_{-0.1265}$
$\text{jitter}_{\text{TESS}}$ Jitter (TESS)	0.02881	$^{+0.0069}_{-0.0061}$
$\text{jitter}_{\text{TASTE}}$ Jitter (TASTE)	0.0004	$^{+0.0002}_{-0.0002}$
b Impact Parameter	0.5732	$^{+0.0144}_{-0.0136}$
R_p/R_j Planet-to-Jupiter Radius Ratio	1.2365	$^{+0.0936}_{-0.0852}$

Table 4: Parameter estimated using MCMC sampling

2013 where the value is 0.1435 ± 0.0008 [17].Our estimation for the semi-major axis (a/R_p) came out to be $6.2715^{+0.1086}_{-0.1128}$, which agrees well with Collins et al. 2017 [3] and Covino et al. 2013 [4] with a value of $6.247^{+0.067}_{-0.065}$ and $6.246^{+0.099}_{-0.096}$, respectively. The inclination came out to be $84.0887°^{+0.1023}_{-0.1202}$ agrees with most of the papers, including those by Collins et al. 2017 [3].

In practice, the limb darkening parameters are unique to each telescope and filter, so exploring the similarities among the results would be constrained. The coefficients were useful simply for modelling the expected shape of the light curve and obtaining other parameters. Another important parameter is the radius of the planet in terms of the radius of Jupiter was estimated to be $1.2365^{+0.0937}_{-0.0853}$ agreeing well with the results from Alsubai et al. 2011 (1.164 ± 0.045) [2] and agreeing that the planet is a Hot Jupiter-like planet. Also, the small value of the semi-major axis, the short orbital period and the large planetary radius indicate that Qatar 1b is a hot Jupiter.

The results could be extended over to multi-epoch observation with TASTE to study if there are any transit timing variations, as well as their spectral properties. Our results agree with Collins et al. [3], which mentioned no other planet in the system, but a longer observation period would be needed to disprove the claim.

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